



Department of Electrical & Computer Engineering
Rajshahi University of Engineering & Technology

Course Plan / Outline

1. **Course Information:**

Course Title	Electrical Machines-I
Course Code	ECE 2107
Course Type	Program Course (Theory)
Credit	3.00 (Three Contact Hours per Week), three lecture slots per week according to class routine.
Semester	2 nd Year Odd Semester

2. **Instructor Information:**

Name of the Instructor	To be announced before starting of the semester
Designation	
Office Room	
Cell	
Email	
Webpage link	

3. **Prerequisite (if have any): None**

4. **Course Rationale:**

Students of the Electrical & Computer Engineering department need to study Electrical Machines-I to be familiar with the intuitive concepts of the fundamental electromagnetic laws that control the operation of electrical machinery such as transformers, generators, and motors. Furthermore, deep insight relating to construction, modeling, analysis, detailed working and modern-day applications of mentioned electrical machines will be carried out so that the students will be able to cope with different industrial machineries.

5. **Course Objectives (COs):**

At the end of this course, the students will

1. Understand the intuitive concepts of the fundamental electromagnetic laws.
2. Understand the construction, operation of single phase and three phase transformers as well as induction motors (IMs).
3. Be able to draw and analyze the equivalent circuits of electrical machineries.
4. Become acquainted with transformer vector group, phase conversion of transformer and different characteristics of transformers and IMs.
5. Be able to test, design and control electrical machineries.

6. **Course Content / Course Description:**

Transformer: Ideal transformer- transformation ratio, no-load and load vector diagrams; actual transformer-equivalent circuit, regulation, short circuit and open circuit tests. Three phase transformer and its connections; Vector group of three phase transformers; Phase conversion.

Three Phase Induction Motor: Rotating magnetic field, equivalent circuit, vector diagram, torque-speed characteristics, effect of changing rotor resistance and reactance on torque-speed curves, motor torque and developed rotor power, no-load test, blocked rotor test, starting and braking and speed control; Induction generator.

Single Phase Induction Motor: Theory of operation, equivalent circuit and starting.

7. **Course Learning Outcomes (CLOs):**

CLO No.	Course Learning Outcomes (CLO)	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Understand and Explain the principles of operation and the basic components of transformers and induction motors.	C2											
CLO2	Analyze transformer characteristics using industry-standard testing protocols.		C4										
CLO3	Analyze the three-phase and single-phase induction motors' construction, characteristics, and equivalent circuits.		C4										

8. **CLOs Assessment Policy:**

CLOs	Class Test	Assignment	Project	Presentation	Final Exam	Others
CLO1	√				√	
CLO2	√				√	
CLO3	√			√	√	

09. **References:**

SL No.	Text Book/Reference Book(s)	Author(s)	Online Availability
1.	Electric Machinery Fundamentals, The McGraw-Hill Companies	Stephen J. Chapman	Link is provided in Google Class Room
2.	Electrical Machinery; KHANNA PUBLISHERS.	Dr. P S Bimbhra	Link is provided in Google Class Room
3.	Generalized Theory of Electrical Machines, KHANNA PUBLISHERS.	Dr. P S Bimbhra	Link is provided in Google Class Room
4.	A Text Book of Electrical Technology (Volume-II), S. Chand PUBLISHERS.	B. L. Theraja, A.K. Theraja	Link is provided in Google Class Room
5.	Direct and alternating current machinery	Jack Rosenblatt, M. Harold Friedman	Link is provided in Google Class Room

10. Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning assessment strategy mapped with CLOs

Week	Topic	Delivery Material	Teaching Learning Strategy	Assessment Strategy	Aligned CLOs
1	Introduction to Machinery Principles: Production of magnetic field, Faraday's Law, Flemings Right hand & Left-hand Rules, Lenz's Law etc. Introduction to Transformer: Operating principle of Transformer, Classification of Transformer. Construction of single phase transformer	Digital Multimedia, Lecture Notes, White Board	☒ Lecture ☐ Tutorial ☒ Discussion ☐ Interaction ☐ Audio/Video ☐ Others	☒ Class Test ☐ Assignment ☒ Final Exam ☐ Presentation ☐ Project ☐ Others	CLO1
2	Ideal Transformer: Transformation Ratio, Impedance transformation through ideal transformer, derivation of emf equation of ideal transformer.	Digital Multimedia, Lecture Notes, White Board	☒ Lecture ☐ Tutorial ☒ Discussion ☐ Interaction ☐ Audio/Video ☐ Others	☒ Class Test ☐ Assignment ☒ Final Exam ☐ Presentation ☐ Project ☐ Others	CLO1
3	Transformer Equivalent Circuit: Determining the Exact and Approximate equivalent circuit of a real transformer and discussing different types of loss occurred in transformer.	Digital Multimedia, Lecture Notes, White Board	☒ Lecture ☐ Tutorial ☒ Discussion ☐ Interaction ☐ Audio/Video ☐ Others	☒ Class Test ☐ Assignment ☒ Final Exam ☐ Presentation ☐ Project ☐ Others	CLO1 & CLO2
4	Transformer Phasor Diagram: (i) Transformer Phasor Diagram at no load condition. (ii) Transformer Phasor Diagram at loaded (resistive, inductive & capacitive) condition.	Digital Multimedia, Lecture Notes, White Board	☒ Lecture ☐ Tutorial ☒ Discussion ☐ Interaction ☐ Audio/Video ☐ Others	☒ Class Test ☐ Assignment ☒ Final Exam ☐ Presentation ☐ Project ☐ Others	CLO1 & CLO2
5	Determining the Values of Components in the Transformer Model: Open circuit test and short circuit test of transformer and determining the Values of Components of the Transformer. Determining the voltage regulation of transformer. CT-1	Digital Multimedia, Lecture Notes, White Board	☒ Lecture ☐ Tutorial ☒ Discussion ☐ Interaction ☐ Audio/Video ☐ Others	☒ Class Test ☐ Assignment ☒ Final Exam ☐ Presentation ☐ Project ☐ Others	CLO1 & CLO2
6	Three-Phase Transformer: Introduction to Three phase transformer, Three phase transformer connections. Vector Group of Three-Phase Transformer: i. Group-1(Zero Degree Phase Displacement) ii. Group-2 (180 degree Phase Displacement)	Digital Multimedia, Lecture Notes, White Board	☒ Lecture ☐ Tutorial ☒ Discussion ☐ Interaction ☐ Audio/Video ☐ Others	☒ Class Test ☐ Assignment ☒ Final Exam ☐ Presentation ☐ Project ☐ Others	CLO1 & CLO2
7	Vector Group of Three-Phase Transformer: iii. Group-3(Minus 30 Degree Phase Displacement) iv. Group-4 (Plus 30 Degree Phase Displacement) Phase conversion: The Open-delta (or V-V) Connection, The Open-Wye-Open-Delta Connection, The Scott-T Connection CT-2	Digital Multimedia, Lecture Notes, White Board	☒ Lecture ☐ Tutorial ☒ Discussion ☐ Interaction ☐ Audio/Video ☐ Others	☒ Class Test ☐ Assignment ☒ Final Exam ☐ Presentation ☐ Project ☐ Others	CLO1 & CLO2
8	Three Phase Induction Motor (IM): Classification of Alternating Current Motors. Construction of three phase IM. Concept of generation of rotating magnetic field and principle of operation of three phase IM. Equivalent Circuit and Vector diagram of three phase IM.	Digital Multimedia, Lecture Notes, White Board	☒ Lecture ☐ Tutorial ☒ Discussion ☐ Interaction ☐ Audio/Video	☒ Class Test ☐ Assignment ☒ Final Exam ☐ Presentation ☐ Project	CLO1 & CLO3

			☐ Others	☐ Others	
9	Three Phase Induction Motor (IM): Relationship between torque and rotor power factor (rotor resistance and reactance), Conditions for maximum torque under starting and running conditions CT-3	Digital Multimedia, Lecture Notes, White Board	☒ Lecture ☐ Tutorial ☒ Discussion ☐ Interaction ☐ Audio/Video ☐ Others	☒ Class Test ☐ Assignment ☒ Final Exam ☐ Presentation ☐ Project ☐ Others	CLO1 & CLO3
10	Three Phase Induction Motor (IM): Torque- Speed characteristics and Effect of changing rotor resistance and reactance torque-speed curves, Power losses in three phase IM, Analysis of motor torque and developed rotor power.	Digital Multimedia, Lecture Notes, White Board	☒ Lecture ☐ Tutorial ☒ Discussion ☐ Interaction ☐ Audio/Video ☐ Others	☒ Class Test ☐ Assignment ☒ Final Exam ☐ Presentation ☐ Project ☐ Others	CLO1 & CLO3
11	Three Phase Induction Motor (IM): No-load test, blocked rotor test of three phase IM, Starting and braking of three phase IM CT-4	Digital Multimedia, Lecture Notes, White Board	☒ Lecture ☐ Tutorial ☒ Discussion ☐ Interaction ☐ Audio/Video ☐ Others	☒ Class Test ☐ Assignment ☒ Final Exam ☒ Presentation ☐ Project ☐ Others	CLO3
12	Three Phase Induction Motor (IM): Methods of Speed controlling of three phase IM Three Phase Induction Motor (IM): Induction generator and its operation Single Phase Induction Motor (IM): Theory of operation of single-phase IM	Digital Multimedia, Lecture Notes, White Board	☒ Lecture ☐ Tutorial ☒ Discussion ☐ Interaction ☐ Audio/Video ☐ Others	☒ Class Test ☐ Assignment ☒ Final Exam ☐ Presentation ☐ Project ☐ Others	CLO1
13	Single Phase Induction Motor (IM): Classification and equivalent circuit of single phase IM, Starting of single phase IM	Digital Multimedia, Lecture Notes, White Board	☒ Lecture ☐ Tutorial ☒ Discussion ☐ Interaction ☐ Audio/Video ☐ Others	☒ Class Test ☐ Assignment ☒ Final Exam ☐ Presentation ☐ Project ☐ Others	CLO1
14	Reserved	-----	-----	-----	-----

11. Virtual Availability of Lecture Materials:

All resources related to this course are available in online platform.

Medium	Google classroom/Zoom/MS Teams etc.
Instructions	
Class Code	

12. Assessment & Marks Distribution:

SL No.	Name of Assessment		Marks	Date
1.	Continuous Assessment	Class Participation and Attendance	10	As Per Class Routine
2.		Class Test	20	As Per Class Test Routine
3.		Assignment/Project/Viva-voce/Presentation etc.	10	Mentioned by the Instructor
4.	Final Examination		60	As Per Examination Routine

Total Marks:	100	
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13. **Grading Policy:**

Mark wise grading policy of this course is as follows:

% of Marks	Letter Grade	Point
80 – 100	A+	4.00
75 – 79	A	3.75
70 – 74	A-	3.50
65 – 69	B+	3.25
60 – 64	B	3.00
55 – 59	B-	2.75
50 – 54	C+	2.50
45 – 49	C	2.25
40 – 44	D	2.00
0 – 39	F	0.00

14. **Additional Course Policy:**

SL. No.	Name of the Test	Description
1.	Class Participation and Attendance	Attending classes are mandatory for the students. Students will be marked in a scale of 10 marks according to Academic Ordinance for the Undergraduate Studies.
2.	Class Test	For this course, four class tests will be held and marks of each class test is 20. Best three class tests average mark is counted as the score of class test. Any unfair means is strictly prohibited in the examination hall.
3.	Assignment/Project/Viva-voce/Presentation etc.	All the assessment strategies are considered as regular assessment. The course teacher will assign works to the students and evaluate them with a scale 10 marks that will be counted as regular assessment marks in final scoring.
4.	Semester Final	For this course, semester final examination will be held according to examination routine at end of this semester. Semester final examination contains 60 marks. Any unfair means is strictly prohibited in the examination hall.

15. **Students Policy with Physically Handicapped:**

Students with disabilities are required to inform the Department of Computer Science & Engineering of any specific requirement for classes or examination as soon as possible

16. **Generic Skills (Indirect):**

SL. No.	Generic skills	Attained Generic Skills (Please tick against the relevant skill)	How the skills will be developed
1.	Presentation skill	√	Presentation
2.	Writing skill		
3.	Interpersonal skill		
4.	Leadership skill		
5.	Community skills		

17. **Quick Links:**

- Website: <https://www.ece.ruet.ac.bd>

